

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

1.

What is the purpose of modifying existing code in the context of training a custom object detector?

1 / 1 point

☒

To adjust code for compatibility with new data sources.

☐

To shorten the training process.

☐

To enhance the performance of the default system libraries.

☐

To improve the aesthetic of your code.

☒

Correct

Correct! Modifying code allows integration of new datasets into the training pipeline.
2.

What is a primary challenge when handling non-standard dataset formats in machine learning projects?

1 / 1 point

☒

The requirement for custom parsing scripts.

☐

The difficulty in visualizing the data.

☐

The inability to perform cross-validation.

☐

The need for additional hardware tools.

☒

Correct

Correct! Non-standard formats often necessitate specific parsing logic to work seamlessly.
3.

Why are simple, non-contextual examples used in learning algorithms?

1 / 1 point

☐

To avoid technical errors

☐

To ensure fast computation

☒

To generalize learning across contexts

☐

To fit examples within time constraints

☒

Correct

Correct! Simplifying examples helps abstract understanding to broader applications.
4.

Which command is used to install TensorFlow using pip?

1 / 1 point

☒

pip install tensorflow

☐

conda install tensorflow

☐

pip install gym

☐

conda update tensorflow

☒

Correct

Correct! 'pip install tensorflow' is the command to install TensorFlow.
5.

Which library is included in Anaconda specifically for Natural Language Processing?

1 / 1 point

☐

Seaborn

☒

NLTK

☐

Pandas

☐

SciKit-learn

☒

Correct

Correct! NLTK is included for NLP tasks.
6.

What is the significance of an epoch in machine learning training?

1 / 1 point

☐

It controls the number of layers in a neural network

☒

It signifies one complete pass over the entire training dataset

☐

It defines the batch size of training data

☐

It measures the learning rate adaptation over time

☒

Correct

Correct! This allows iterative learning and adjustment after dataset review.
7.

What mental skill does coding by oneself develop, according to educational practices?

1 / 1 point

☐

Memory retention

☒

Mental toughness and initiative

☐

Collaborative skills

☐

Time management

☒

Correct

Correct! Coding by oneself enhances these critical mental skills.
8.

What characteristic makes dynamic neural networks challenging for most libraries?

1 / 1 point

☒

Their varying architecture with inputs

☐

Their computational inefficiencies

☐

Their reliance on supervised learning

☐

Their need for large datasets

☒

Correct

Correct! Dynamic networks adapt their structure, posing challenges for static frameworks.
9.

Why is it recommended not to run all code within the same notebook in Google's Colab when experimenting with models?

1 / 1 point

☐

To prevent data loss

☒

To avoid exhausting Colab resources

☐

To improve code readability

☐

To increase computation speed

☒

Correct

Correct! Running extensive code in one notebook can exhaust Colab's resources.
10.

Why is experimenting with different model backbones important in practical applications?

1 / 1 point

☒

It provides insight into how different architectures handle the same data differently

☐

It is required to successfully deploy every machine learning project

☐

It guarantees higher performance compared to fixed backbones

☐

It simplifies the preprocessing pipeline

☒

Correct

Correct! It helps in evaluating architecture suitability for specific datasets or tasks.
11.

What does the Markov assumption imply in model design?

1 / 1 point

☐

All past states affect future states

☒

The current state depends only on the previous state

☐

The current state depends on a fixed constant

☐

Every future event is independent

☒

Correct

Correct! This is a core idea in Markov models.
12.

In neural networks, what does 'upsampling flow' refer to?

1 / 1 point

☒

Increasing spatial dimensions

☐

Ensuring data normalization

☐

Reducing feature map depth

☐

Downgrading model precision

☒

Correct

Correct! Upsampling increases spatial resolution, reversing convolution's compression.